

# Spintronic Methods

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## 1 Notation and Conventions

Unless a section explicitly states otherwise, vectors are bold, unit vectors carry a hat, time is measured in seconds, and magnetic fields  $\mathbf{H}$  are expressed in A m<sup>−1</sup>. The following symbols are used throughout the library.

$\mathbf{m}$  Normalized magnetization, with  $\|\mathbf{m}\| = 1$ .

$\mathbf{s}_i$  Unit spin direction at site  $i$ .

$M_s$  Saturation magnetization of the ferromagnetic layer.

$\mathbf{H}_{\text{eff}}$  Effective magnetic field.

$\boldsymbol{\sigma}$  Spin-polarization unit vector.

$\alpha$  Dimensionless Gilbert damping parameter.

$\gamma$  Positive gyromagnetic ratio,  $\gamma = g\mu_B/\hbar$ , where  $g$  is the Landé  $g$ -factor.

$\gamma_0$  Positive gyromagnetic factor for an  $\mathbf{H}$ -field formulation,  $\gamma_0 = \mu_0 |\gamma|$ .

$\theta_{\text{SH}}$  Spin-Hall angle.

$J_c$  Charge-current density.

$t_{\text{FM}}$  Ferromagnetic-layer thickness.

$\beta$  Signed torque ratio,  $\beta = \tau_{\text{FLT}}/\tau_{\text{DLT}}$ .

$\tau_{\text{DLT}}$ ,  $\tau_{\text{FLT}}$  Damping-like and field-like torque amplitudes, respectively; both have field units.

$e$  Positive elementary charge.

$\mu_0$  Vacuum permeability.

$\mu_B$  Bohr magneton.

$\hbar$  Reduced Planck constant.

$k_B$  Boltzmann constant.

$T$  Absolute temperature.

**Physical constants.** The following values reproduce `constantfile.m`:

$$\begin{aligned}\gamma &= 1.760859644 \times 10^{11} \text{ rad s}^{-1} \text{ T}^{-1}, & e &= 1.602176565 \times 10^{-19} \text{ C}, \\ \mu_0 &= 1.25663706143592 \times 10^{-6} \text{ T m A}^{-1}, & \mu_B &= 9.274 \times 10^{-24} \text{ J T}^{-1}, \\ \hbar &= 6.58211951440 \times 10^{-16} \text{ eV s}, & k_B &= 1.38064852 \times 10^{-23} \text{ J K}^{-1}.\end{aligned}$$

**Convention warning.** Some references define the effective field as  $\mathbf{B}_{\text{eff}}$  in tesla and use  $|\gamma|$ , whereas this library uses  $\mathbf{H}_{\text{eff}}$  in  $\text{A m}^{-1}$  and  $\gamma_0 = \mu_0 |\gamma|$ . Check this convention before copying an equation into simulation code.

## 2 Spintronics Concepts

### 2.1 Antiferromagnets

An antiferromagnet is a magnetic material in which neighboring magnetic moments prefer an antiparallel arrangement. The opposing moments can compensate, giving little or no net magnetization even though the material has long-range magnetic order. Its order is therefore described more naturally by a Néel vector than by the net magnetization alone [9].

### 2.1.1 A-Type Antiferromagnet

An A-type antiferromagnet consists of ferromagnetic planes stacked antiferromagnetically: spins within each plane point in the same direction, all spins in the next plane point in the opposite direction, and the pattern repeats along the stacking direction. The word *plane* refers to the crystallographic planes selected by the exchange interactions in a particular material. Figure 1 chooses the  $z$  axis as the stacking direction as an illustrative convention, not as a universal crystal orientation.

**Source status. Reference needed.** This A-type structural classification was supplied directly for these notes and still needs a dedicated literature reference.

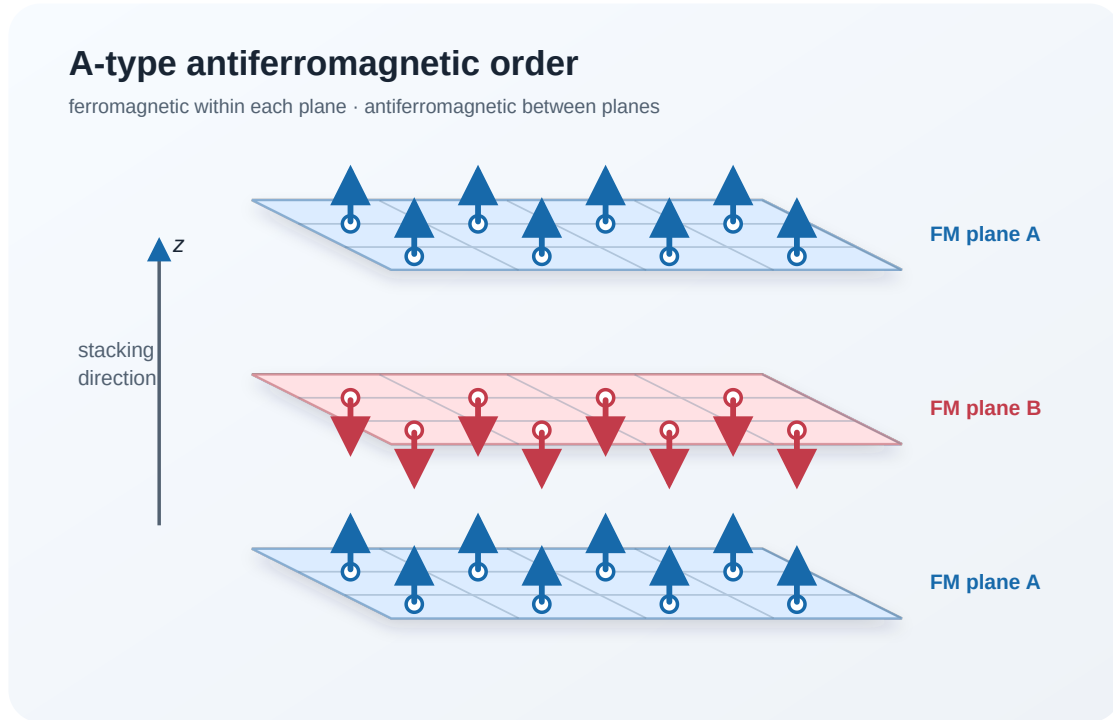


Figure 1: A-type antiferromagnetic stacking: ferromagnetic planes alternate in direction. Original schematic; the physical classification follows the provisional description above and therefore shares its **reference-needed** status.

For classical unit spins  $\mathbf{s}_i$ , the exchange part of the atomistic Hamiltonian can be written in the ASTRO sign convention as [11]

$$\mathcal{H}_{\text{ex}} = \frac{1}{2} \sum_{i \neq j} J_{ij} \mathbf{s}_i \cdot \mathbf{s}_j. \quad (2.1)$$

In this convention, A-type order is favored by

$$J_{\parallel} < 0, \quad J_{\perp} > 0, \quad (2.2)$$

where  $J_{\parallel}$  couples neighbors within a plane and  $J_{\perp}$  couples neighbors in adjacent planes. Thus the in-plane bonds favor parallel alignment, whereas the interplane bonds favor antiparallel alignment. Some publications place a minus sign in front of the exchange sum; their signs of  $J$  are reversed. The sign convention and pair counting in Eq. (2.1) are the same as those used in the full atomistic Hamiltonian in Eq. (5.1) and in ASTRO [11].

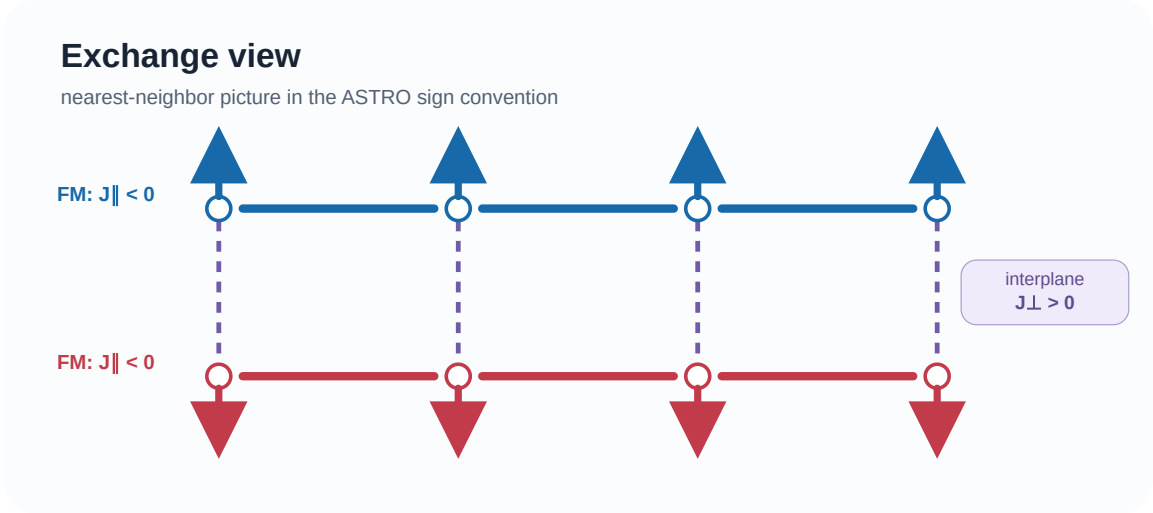


Figure 2: Exchange pattern for A-type order: parallel bonds within a layer and antiparallel bonds between layers. Original schematic based on the cited ASTRO exchange-sign convention [11]; the A-type classification itself remains **reference needed**.

For two equivalent alternating layers with normalized layer magnetizations  $\mathbf{m}_A$  and  $\mathbf{m}_B$ , define the net and Néel order parameters as [9]

$$\mathbf{m} = \frac{\mathbf{m}_A + \mathbf{m}_B}{2}, \quad \mathbf{n} = \frac{\mathbf{m}_A - \mathbf{m}_B}{2}. \quad (2.3)$$

For an ideal compensated A-type state, these definitions give

$$\mathbf{m}_B = -\mathbf{m}_A, \quad \mathbf{m} = \mathbf{0}, \quad |\mathbf{n}| = 1. \quad (2.4)$$

The defining feature is therefore not simply zero magnetization. It is the specific spatial pattern of ferromagnetic order in two directions and antiferromagnetic alternation in the third. This distinguishes A-type order from G-type antiferromagnetism, in which every nearest neighbor is antiparallel, and C-type order, in which ferromagnetic chains are coupled antiferromagnetically. **Reference needed** for this final classification comparison.

### 3 Symmetries in Spin Systems

A symmetry operation transforms the spatial coordinates, momenta, spins, and external fields of a system. It is a symmetry of a Hamiltonian only when the transformed Hamiltonian describes the same physical system. For a unitary operation  $G$  this condition is

$$G\mathcal{H}G^{-1} = \mathcal{H}, \quad (3.1)$$

whereas time reversal is antiunitary and also complex-conjugates scalar coefficients.

**Source status. Reference needed.** The general time-reversal and spatial-inversion operator rules in this section are included provisionally because no approved general reference is currently available. The mirror rules are supported by the EE237 notes, the isotropic-exchange example by ASTRO, and the magnetic parity–time example by Zhang *et al.*, as cited locally below.

### 3.1 Polar and Axial Vectors

Position  $\mathbf{r}$ , momentum  $\mathbf{p}$ , electric field  $\mathbf{E}$ , and charge-current density  $\mathbf{J}$  are polar vectors. Orbital angular momentum  $\mathbf{L} = \mathbf{r} \times \mathbf{p}$ , spin  $\mathbf{S}$ , magnetization  $\mathbf{M}$ , and magnetic field  $\mathbf{B}$  are axial vectors. For an orthogonal spatial transformation represented by  $R$ , the two vector classes transform as

$$\mathbf{v}' = R\mathbf{v}, \quad \mathbf{a}' = (\det R)R\mathbf{a}, \quad (3.2)$$

where  $\mathbf{v}$  is polar and  $\mathbf{a}$  is axial. Thus they transform in the same way under proper rotations ( $\det R = +1$ ) and differently under mirrors and inversion ( $\det R = -1$ ) [20].

The most frequently used transformations are summarized below. The entries describe the vector attached to the spatial point after that point has been mapped by the operation.

| Operation                         | $\mathbf{r}, \mathbf{p}$                                                       | $\mathbf{S}, \mathbf{M}$                                                        | $\mathbf{E}, \mathbf{B}$                                                |
|-----------------------------------|--------------------------------------------------------------------------------|---------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| Time reversal<br>$\mathcal{T}$    | $\mathbf{r} \rightarrow \mathbf{r}, \mathbf{p} \rightarrow -\mathbf{p}$        | $\mathbf{S}, \mathbf{M} \rightarrow -\mathbf{S}, -\mathbf{M}$                   | $\mathbf{E} \rightarrow \mathbf{E}, \mathbf{B} \rightarrow -\mathbf{B}$ |
| Inversion $\mathcal{P}$           | $\mathbf{r}, \mathbf{p} \rightarrow -\mathbf{r}, -\mathbf{p}$                  | $\mathbf{S}, \mathbf{M} \rightarrow \mathbf{S}, \mathbf{M}$                     | $\mathbf{E} \rightarrow -\mathbf{E}, \mathbf{B} \rightarrow \mathbf{B}$ |
| Mirror $\mathcal{M}_{\mathbf{n}}$ | $\mathbf{v} \rightarrow \mathbf{v} - 2\mathbf{n}(\mathbf{n} \cdot \mathbf{v})$ | $\mathbf{a} \rightarrow -\mathbf{a} + 2\mathbf{n}(\mathbf{n} \cdot \mathbf{a})$ | polar $\mathbf{E}$ ; axial $\mathbf{B}$                                 |

### 3.2 Time-Reversal Symmetry

Time reversal changes  $t \rightarrow -t$  and reverses momenta and angular momenta. Spin is an intrinsic angular momentum, so

$$\mathcal{T}\mathbf{S}\mathcal{T}^{-1} = -\mathbf{S}, \quad \mathcal{T}i\mathcal{T}^{-1} = -i. \quad (3.3)$$

For a classical magnetization trajectory, the corresponding transformed trajectory is

$$\mathbf{m}_{\mathcal{T}}(t) = -\mathbf{m}(-t). \quad (3.4)$$

A magnetic state can therefore break time-reversal symmetry even when its nonmagnetic crystal structure does not. A fixed applied magnetic field, a fixed current, damping, or another nonequilibrium drive must not be silently left unchanged when testing time reversal; these time-odd or irreversible ingredients generally prevent the driven equation from being time-reversal-invariant.

**Source status. Reference needed.** Equations (3.3) and (3.4), and the general invariance statement above, still need an approved reference.

### 3.3 Parity or Spatial-Inversion Symmetry

Here parity means full spatial inversion,

$$\mathcal{P} : \quad \mathbf{r} \rightarrow -\mathbf{r}. \quad (3.5)$$

Because spin is axial, inversion moves a spin to the inverted site without reversing its direction. For a spin texture this gives

$$\mathbf{S}_{\mathcal{P}}(\mathbf{r}) = \mathbf{S}(-\mathbf{r}). \quad (3.6)$$

The structure has inversion symmetry only when the inverted sites, chemical species, interactions, and spin pattern reproduce the original structure. This is distinct from particle-exchange symmetry. In non-Hermitian coupled-mode models the symbol  $\mathcal{P}$  may instead denote exchange of two subsystems; its meaning must therefore be stated explicitly.

**Source status. Reference needed.** Equations (3.5) and (3.6), including the axial-spin convention under inversion, still need an approved general reference.

### 3.4 Mirror Symmetry

A mirror with unit normal  $\mathbf{n}$  reverses the perpendicular component of a polar vector. An axial vector behaves oppositely: its components parallel to the mirror plane reverse, while its perpendicular component is unchanged according to Eq. (3.2) [20]. For the  $xy$  mirror,

$$(v_x, v_y, v_z) \rightarrow (v_x, v_y, -v_z), \quad (S_x, S_y, S_z) \rightarrow (-S_x, -S_y, S_z). \quad (3.7)$$

The transformed lattice and every axial spin must both be compared with the original magnetic structure. A lattice may therefore possess a mirror plane that is broken by its magnetic order.

### 3.5 Spin-Rotation Symmetry

Spatial rotations move the lattice, whereas spin rotations act in spin space. For the isotropic exchange Hamiltonian used in ASTRO,

$$\mathcal{H}_{\text{ex}} = \frac{1}{2} \sum_{i \neq j} J_{ij} \mathbf{S}_i \cdot \mathbf{S}_j, \quad (3.8)$$

a common spin rotation  $R_s$  leaves every dot product unchanged:

$$(R_s \mathbf{S}_i) \cdot (R_s \mathbf{S}_j) = \mathbf{S}_i \cdot \mathbf{S}_j. \quad (3.9)$$

The exchange term consequently has global spin-rotation symmetry when the couplings  $J_{ij}$  do not depend on spin direction [11]. Magnetic anisotropy, Dzyaloshinskii–Moriya interaction, dipolar coupling, an applied field, or another spin–orbit-coupled term can reduce this continuous symmetry to the subset of joint spatial and spin operations that preserves the complete Hamiltonian.

### 3.6 Combined Parity–Time Symmetry in a Two-Sublattice Magnet

For the two-sublattice order parameters defined in Eq. (2.3), suppose that inversion exchanges sublattices  $A$  and  $B$ . Since inversion does not reverse an axial spin, whereas time reversal does, their actions are

|                |               |               |
|----------------|---------------|---------------|
|                | $\mathbf{m}$  | $\mathbf{n}$  |
| $\mathcal{P}$  | $\mathbf{m}$  | $-\mathbf{n}$ |
| $\mathcal{T}$  | $-\mathbf{m}$ | $-\mathbf{n}$ |
| $\mathcal{PT}$ | $-\mathbf{m}$ | $\mathbf{n}$  |

(3.10)

Thus a compensated antiferromagnetic pattern can be invariant under the combined operation even when it is not invariant under  $\mathcal{P}$  or  $\mathcal{T}$  separately. In a different use of parity, Zhang *et al.* define  $\mathcal{P}$  as interchange of two coupled ferrimagnetic layers and obtain parity–time-symmetric Néel-vector dynamics at angular-momentum compensation [16]. These two meanings should not be conflated.

**Source status. Reference needed.** Equation (3.10) is a direct derivation from the provisional general transformation rules above. The cited paper supports only the separate coupled-layer example.

### 3.7 Practical Symmetry Test

To test a proposed symmetry of a spin model:

1. map every position and sublattice label;
2. transform spin as an axial vector and reverse it additionally under time reversal;

3. transform external fields, currents, and boundary conditions;
4. rebuild every Hamiltonian or dynamical term with the transformed quantities; and
5. compare the complete transformed model and magnetic pattern with the originals.

Checking only the atomic lattice or only the spin arrows is insufficient for a magnetic symmetry.

**Source status. Reference needed.** The general five-step test is a provisional pedagogical synthesis and still needs an approved reference.

## 4 Representative Material Parameter Sets

The values below reproduce particular numerical models from papers and repository implementations authored or coauthored by Zhifeng Zhu. They are model-specific parameter sets rather than universal material constants. In particular, quantities associated with a heavy-metal layer, spin polarization, device geometry, or numerical integration are identified separately from intrinsic magnetic parameters. Fixed-length unit vectors are used in all three source-model families summarized here.

### 4.1 Property of Heavy Metal

The spin-Hall angle belongs to the heavy-metal spin-current source and is not specific to a ferromagnetic, ferrimagnetic, or antiferromagnetic target layer. For the signed representative values in Table 1, we use  $\mathbf{J}_s = \theta_{\text{SH}} \boldsymbol{\sigma} \times \mathbf{J}_c$  and choose Pt as the positive reference; Ta and W then have negative signs [5, 6, 7]. Reversing the current, layer-normal, or spin-polarization convention requires a corresponding sign change in the torque implementation.

| Property                                     | Value                                                    | Evidence and qualification                                                                                                |
|----------------------------------------------|----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Pt spin-Hall angle                           | +0.07                                                    | Representative literature value quoted by Liu et al. and Pai et al.; defines the positive reference sign                  |
| High-resistivity $\beta$ -Ta spin-Hall angle | −0.12 to −0.15                                           | Consistent magnitudes from ST-FMR, perpendicular switching, and three-terminal switching                                  |
| High-resistivity $\beta$ -W spin-Hall angle  | −0.30±0.02 (ST-FMR); −0.33±0.06 (switching)              | Negative sign verified relative to Pt; the ST-FMR value is a lower bound if spin diffusion suppresses the measured torque |
| Heavy-metal resistivity                      | $\rho_{\text{HM}} = 200 \times 10^{-8} \Omega \text{ m}$ | Representative Ta/W value from the compact model; $\rho_{\text{HM}} = 200 \mu\Omega \text{ cm}$                           |

Table 1: Representative properties of Pt,  $\beta$ -Ta, and  $\beta$ -W heavy metals [5, 6, 7].

These values are not universal elemental constants. In particular, Pai et al. find that W changes strongly with film phase and resistivity: a mixed  $\alpha/\beta$  film gives  $|\theta_{\text{SH}}| = 0.18 \pm 0.02$ , whereas their predominantly  $\alpha$ -W film gives only the upper bound  $|\theta_{\text{SH}}| < 0.07$  [7]. The material-specific tables below omit  $\theta_{\text{SH}}$ ; select a value from the heavy-metal source and sign convention used by the model.

The compact-model README compares reported resistivities of  $190 \mu\Omega \text{ cm}$  for Ta and  $170$ – $260 \mu\Omega \text{ cm}$  for W, then uses the representative value  $\rho_{\text{HM}} = 200 \mu\Omega \text{ cm}$  [5].

## 4.2 Ferromagnet: CoFeB

The CoFeB free layer is represented by a one-sublattice macrospin LLGS model with perpendicular anisotropy. The anisotropy field and saturation magnetization below are the defaults of the SOT/STT compact model; the remaining magnetic and drive parameters come from the model of Zhang et al. [5, 4]. The compact-model parameter named  $\mathbb{H}\mathbf{k}$  is a flux-density quantity in tesla. Its corresponding  $\mathbf{H}$  field in  $\text{A m}^{-1}$  is obtained by dividing by  $\mu_0$ .

| Quantity                  | Value                            | Role or convention                                                                               |
|---------------------------|----------------------------------|--------------------------------------------------------------------------------------------------|
| Uniaxial anisotropy field | $B_k = 1.5 \text{ T}$            | Compact-model parameter $\mathbb{H}\mathbf{k}$ ; $H_k \approx 1.19 \times 10^6 \text{ A m}^{-1}$ |
| Saturation magnetization  | $M_s = 1000 \text{ emu cm}^{-3}$ | $M_s = 1.00 \times 10^6 \text{ A m}^{-1}$ ; $\mu_0 M_s \approx 1.26 \text{ T}$                   |
| Gilbert damping           | $\alpha = 0.02$                  | Magnetic model parameter                                                                         |

Table 2: Selected CoFeB macrospin parameters from the SOT/STT compact model and Zhang et al. [5, 4].

**Parameter justification.** Based on Nat. Mater. **9**, 721 (2010).,  $H_{\text{an}} = 2(K_{\text{bulk}} + K_i/t_{\text{FL}})/M_s$  with  $K_{\text{bulk}} = 2.245 \times 10^5 \text{ J m}^{-3}$ ,  $K_i = 1.286 \times 10^{-3} \text{ J m}^{-2}$ ,  $M_s = 1.58 \text{ T} = 1257 \text{ emu cm}^{-3}$ , when we use  $t_{\text{FL}} = 1.2 \text{ nm}$ , it gives  $H_{\text{an}} = 2.06 \text{ T}$ . Here we approximate  $M_s = 1000 \text{ emu cm}^{-3}$ ,  $H_k = 1.5 \text{ T}$  [5].

## 4.3 Ferrimagnet: GdFeCo

Zhang et al. use a random two-dimensional atomistic  $(\text{FeCo})_{1-x}\text{Gd}_x$  lattice driven by damping-like STT, whereas Zhu et al. use a one-dimensional atomistic GdFeCo model driven by SOT [14, 15]. Both are fixed-length LLG or LLGS models. The 2022 paper states that its parameters are otherwise the same as those in the 2020 paper, but replaces the single exchange coefficient with three bond-specific values. Table 3 therefore keeps the two published exchange parameterizations separately rather than selecting or averaging them.

The exchange values are deliberately not converted into a single common coefficient. The 2020 value belongs to a one-dimensional nearest-neighbor Hamiltonian, while the 2022 values distinguish the chemical identity of each bond in a random two-dimensional alloy. Their lattice topology and bond statistics must be retained with the numerical values when either set is used.

## 4.4 Noncollinear Antiferromagnet: $\text{Mn}_3\text{Sn}$

The three-sublattice macrospin switching model of Xu et al. and the later spatially resolved atomistic domain-wall model use the same magnetic moment, DMI, anisotropy, damping, and spin-Hall angle, but different exchange coefficients [8, 10]. The values must therefore remain attached to their model and interaction convention. Both Hamiltonians take positive  $A$  to favor antiparallel alignment, and the crystalline easy axes point toward the nearest Sn atoms. The atomistic domain-wall paper additionally states that its positive  $D$  favors clockwise order.

The  $\text{Mn}_3\text{Sn}$  entries are not by themselves a complete crystallographic simulation specification: a reproducible atomistic calculation must also retain the source model’s lattice, site mapping, bond list, DMI-vector orientations, boundary conditions, and pair-counting convention.

| Quantity                  | Zhu et al. (2020)                                                                                                                      | Zhang et al. (2022)                                                                                                                                  |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Spatial model             | 1D atomistic lattice with lattice constant $d = 0.4$ nm; strip length and width are 400 and 40 nm                                      | Random 2D square lattice; each interior site has four nearest neighbors; the main phase diagrams use 100 atoms                                       |
| Exchange                  | $A_{\text{ex}} = +3$ meV = $+4.8065 \times 10^{-22}$ J                                                                                 | $A_{\text{Gd-Gd}} = -1.26 \times 10^{-21}$ J;<br>$A_{\text{FeCo-FeCo}} = -2.83 \times 10^{-21}$ J;<br>$A_{\text{FeCo-Gd}} = +1.09 \times 10^{-21}$ J |
| Perpendicular anisotropy  | $K_{\text{TM}} = K_{\text{RE}} = 0.04$ meV                                                                                             | same as Zhu et al. (2020)                                                                                                                            |
| Hard-axis anisotropy      | $\kappa_{\text{TM}} = \kappa_{\text{RE}} = 0.2$ $\mu\text{eV}$                                                                         | Not included in the stated Hamiltonian                                                                                                               |
| DMI                       | $D = 0.128$ meV                                                                                                                        | Not included in the stated Hamiltonian                                                                                                               |
| Gilbert damping           | $\alpha_{\text{TM}} = \alpha_{\text{RE}} = 0.015$                                                                                      | same as Zhu et al. (2020)                                                                                                                            |
| Landé factors             | $g_{\text{TM}} = 2.2$ ; $g_{\text{RE}} = 2.0$                                                                                          | same as Zhu et al. (2020)                                                                                                                            |
| Sublattice magnetizations | $M_{s,\text{TM}} = 1149$ kA m $^{-1}$ ;<br>$M_{s,\text{RE}} = 1012$ kA m $^{-1}$ at $T = 300$ K                                        | same as Zhu et al. (2020)                                                                                                                            |
| Ferrimagnet thickness     | $t_{\text{FiM}} = 0.4$ nm                                                                                                              | same as Zhu et al. (2020)                                                                                                                            |
| Drive parameter           | Spin-Hall angle $\theta_{\text{SH}} = 0.2$ ; field-like to damping-like ratio $\beta$ is varied rather than fixed in the baseline list | STT efficiency $\eta$ is introduced but no numerical value is stated                                                                                 |
| Compensation composition  | Not stated as a composition in the parameter block                                                                                     | $x_{\text{MC}} = 0.23$ ; $x_{\text{AMC}} = 0.21$                                                                                                     |

Table 3: Atomistic GdFeCo parameter sets retained in their native model conventions [15, 14].

| Quantity                  | Value                              | Model scope or convention                                                          |
|---------------------------|------------------------------------|------------------------------------------------------------------------------------|
| Mn moment                 | $m_s = 3\mu_B$                     | Per Mn site in both models                                                         |
| Nearest-neighbor exchange | $A = 17.53$ meV                    | Three-sublattice macrospin switching model                                         |
| Nearest-neighbor exchange | $A = 9$ meV                        | Spatial atomistic domain-wall model                                                |
| DMI coefficient           | $D = 0.833$ meV                    | Same-layer interaction in both models                                              |
| Crystalline anisotropy    | $K = 0.196$ meV                    | Single-site Hamiltonian coefficient in both models                                 |
| Gilbert damping           | $\alpha = 0.003$                   | Same in both fixed-length models                                                   |
| Unit-cell volume          | $V_{\text{cell}} = 0.3778$ nm $^3$ | Macrospin model; $M_s = 6m_s/V_{\text{cell}} \approx 4.42 \times 10^5$ A m $^{-1}$ |
| Mn $_3$ Sn thickness      | $d = 30$ nm                        | Macrospin SOT model geometry                                                       |
| Time step                 | $\Delta t = 5$ fs                  | Fourth-order Runge-Kutta macrospin integration                                     |
| Nanoribbon length         | $5.7$ $\mu\text{m}$                | Spatial atomistic domain-wall model geometry                                       |

Table 4: Mn $_3$ Sn parameter sets used in the macrospin switching and atomistic domain-wall models [8, 10].

## 5 Spin Hamiltonians

### 5.1 Atomistic Spin Hamiltonian

**Equation.** Following the ASTRO exchange-sign convention, a classical spin Hamiltonian containing exchange, uniaxial anisotropy, Dzyaloshinskii–Moriya interaction, and Zeeman energy is

$$\begin{aligned} \mathcal{H} = & + \frac{1}{2} \sum_{i \neq j} J_{ij} \mathbf{s}_i \cdot \mathbf{s}_j - \sum_i K_i (\mathbf{s}_i \cdot \hat{\mathbf{e}}_i)^2 \\ & - \frac{1}{2} \sum_{i \neq j} \mathbf{D}_{ij} \cdot (\mathbf{s}_i \times \mathbf{s}_j) - \sum_i \boldsymbol{\mu}_i \cdot \mathbf{B}_{\text{ext}}. \end{aligned} \quad (5.1)$$

**Definitions.**  $J_{ij}$  is the exchange-link energy in the ASTRO sign convention:  $J_{ij} < 0$  favors parallel alignment, whereas  $J_{ij} > 0$  favors antiparallel alignment.  $K_i$  is the uniaxial anisotropy energy,  $\hat{\mathbf{e}}_i$  is the easy-axis direction,  $\mathbf{D}_{ij}$  is the Dzyaloshinskii–Moriya vector, and  $\boldsymbol{\mu}_i$  is the magnetic moment at site  $i$ .

**Assumptions.** Each  $\mathbf{s}_i$  is a classical unit vector. The pair coefficients satisfy  $J_{ji} = J_{ij}$  and  $\mathbf{D}_{ji} = -\mathbf{D}_{ij}$ . The factors of 1/2 assume that the pair sums include both ordered pairs  $(i, j)$  and  $(j, i)$ . If each physical bond is instead included only once, for example in a unique-bond sum  $\sum_{\langle i, j \rangle}$ , both factors of 1/2 must be omitted. These are equivalent pair-counting conventions and must not be mixed.

**Pair counting and effective field.** Let  $\mu_i = |\boldsymbol{\mu}_i|$  and define the effective flux density by

$$\mathbf{B}_{\text{eff},i} = -\frac{1}{\mu_i} \frac{\partial \mathcal{H}}{\partial \mathbf{s}_i}. \quad (5.2)$$

For the ordered-pair Hamiltonian above, the exchange and DMI contributions are

$$\mathbf{B}_{\text{ex},i} = -\frac{1}{\mu_i} \sum_{j \neq i} J_{ij} \mathbf{s}_j, \quad \mathbf{B}_{\text{DMI},i} = \frac{1}{\mu_i} \sum_{j \neq i} \mathbf{s}_j \times \mathbf{D}_{ij}. \quad (5.3)$$

The two ordered occurrences of each bond give identical derivatives, which cancel the Hamiltonian factors of 1/2. Consequently, no extra factor of 1/2 belongs in either effective-field expression.

**Units.** All terms in  $\mathcal{H}$  have units of energy. Here the external flux density  $\mathbf{B}_{\text{ext}}$  is measured in tesla, so no additional factor of  $\mu_0$  appears in the Zeeman term.

**Source.** For the extended classical spin Hamiltonian and Hamiltonian-derivative field convention, see Evans et al. [2].

**Implementation notes.** The ASTRO implementation evaluates the field,  $\mathbf{B}_{\text{eff},i} = -(1/\mu_i) \partial \mathcal{H} / \partial \mathbf{s}_i$ , rather than the total energy and includes both forward and backward nearest neighbors. The  $J_{ij}$  used above is therefore the coefficient stored in ASTRO’s exchange-link arrays: its exchange field is  $-(1/\mu_i) \sum_j J_{ij} \mathbf{s}_j$  [11]. No factor of 1/2 should appear in the exchange or DMI field formulas because differentiation produces two identical bond contributions, cancelling the Hamiltonian factor. ASTRO’s DMI component expressions correspond to  $\mathbf{D}_{ij} = D(\hat{\mathbf{z}} \times \hat{\mathbf{r}}_{ij})$  and the antisymmetry  $\mathbf{D}_{ji} = -\mathbf{D}_{ij}$  used above.

## 5.2 Macrospin Effective Fields

**Field convention.** For a fixed-length macrospin with normalized magnetization  $\mathbf{m}$ , let  $w$  denote the magnetic energy density in  $\text{J m}^{-3}$ . The effective magnetic field and the corresponding effective flux density are [18]

$$\mathbf{H}_{\text{eff}} = -\frac{1}{\mu_0 M_s} \frac{\partial w}{\partial \mathbf{m}}, \quad \mathbf{B}_{\text{eff}} = \mu_0 \mathbf{H}_{\text{eff}}. \quad (5.4)$$

Thus, an LLG equation written with  $\gamma_0 = \mu_0 |\gamma|$  uses  $\mathbf{H}_{\text{eff}}$  in  $\text{A m}^{-1}$ , whereas one written with  $|\gamma|$  uses  $\mathbf{B}_{\text{eff}}$  in tesla. The two conventions are physically equivalent but their field variables must not be mixed.

Unless stated otherwise,  $K_u$  and  $A_{12}$  are energy densities in  $\text{J m}^{-3}$ . The interfacial  $K_i$  in the combination  $K_{\text{bulk}} + K_i/t_{\text{FM}}$  is instead in  $\text{J m}^{-2}$ , and every saturation magnetization is in  $\text{A m}^{-1}$ . The tensors  $\mathbf{N}$  and  $\mathbf{K}_{fp}$  are dimensionless.

### 5.2.1 One-Sublattice Model

**Energy density and field decomposition.** For uniaxial anisotropy with easy axis  $\hat{\mathbf{u}}$ , a dimensionless symmetric demagnetization tensor  $\mathbf{N}$ , an applied field  $\mathbf{H}_{\text{ext}}$ , and a dipolar field  $\mathbf{H}_{\text{dip}}$  produced by another magnetic body, use the uniaxial-anisotropy convention from slides 13–14 of the EE237 Week 5 lecture notes [18] and take

$$w_1 = K_u \left[ 1 - (\mathbf{m} \cdot \hat{\mathbf{u}})^2 \right] + \frac{\mu_0 M_s^2}{2} \mathbf{m} \cdot \mathbf{N} \mathbf{m} - \mu_0 M_s \mathbf{m} \cdot (\mathbf{H}_{\text{ext}} + \mathbf{H}_{\text{dip}}). \quad (5.5)$$

**External field.** The part of the final term in Eq. (5.5) that is proportional to  $\mathbf{H}_{\text{ext}}$  is the Zeeman coupling to the applied field. Its derivative contributes  $\mathbf{H}_{\text{ext}}$  directly to  $\mathbf{H}_{\text{eff}}$  [18].

**Uniaxial anisotropy field.** Equation (5.4) gives [18]

$$\mathbf{H}_{\text{an}} = \frac{2K_u}{\mu_0 M_s} (\mathbf{m} \cdot \hat{\mathbf{u}}) \hat{\mathbf{u}}, \quad \mathbf{B}_{\text{an}} = \frac{2K_u}{M_s} (\mathbf{m} \cdot \hat{\mathbf{u}}) \hat{\mathbf{u}}. \quad (5.6)$$

For  $\hat{\mathbf{u}} = \hat{\mathbf{z}}$ , the anisotropy term reduces to  $K_u(1 - m_z^2) = K_u \sin^2 \theta$ , as on slide 13. Near  $m_z = 1$ , slide 14 therefore gives  $H_{\text{an},z} = 2K_u/(\mu_0 M_s)$ . For a thin film with bulk and interfacial uniaxial anisotropy,  $K_u = K_{\text{bulk}} + K_i/t_{\text{FM}}$ .

**Demagnetization field.** Slide 21 of the same notes writes the demagnetization self-energy as [18]

$$E_{\text{demag}} = -\frac{\mu_0}{2} \int_V \mathbf{M}(\mathbf{r}) \cdot \mathbf{H}_{\text{demag}}(\mathbf{r}) \, d^3r, \quad (5.7)$$

$$w_{\text{demag}} = -\frac{\mu_0}{2} \mathbf{M} \cdot \mathbf{H}_{\text{demag}} = \frac{\mu_0 M_s^2}{2} \mathbf{m} \cdot \mathbf{N} \mathbf{m} \quad (\text{uniform macrospin}). \quad (5.8)$$

The factor 1/2 avoids double counting the magnet's self-interaction. Equation (5.4) gives

$$\mathbf{H}_{\text{demag}} = -M_s \mathbf{N} \mathbf{m}, \quad \mathbf{B}_{\text{demag}} = -\mu_0 M_s \mathbf{N} \mathbf{m}. \quad (5.9)$$

For principal axes,  $\mathbf{N} = \text{diag}(N_x, N_y, N_z)$  and  $N_x + N_y + N_z = 1$ .

**Dipolar field.** Let a pinned macrospin have volume  $V_p$ , saturation magnetization  $M_{s,p}$ , and direction  $\mathbf{m}_p$ , so its magnetic moment is  $\boldsymbol{\mu}_p = V_p M_{s,p} \mathbf{m}_p$ . In the point-dipole approximation, at displacement  $\mathbf{r} = r \hat{\mathbf{r}}$  from the pinned layer to the free layer, slide 22 gives the dipole formula [18]. The slide denotes it by  $\mathbf{H}_i$ , but its factor  $\mu_0$  makes it a flux density in tesla; therefore, in the convention of this document,

$$\mathbf{B}_{\text{dip}} = \frac{\mu_0}{4\pi r^3} [3(\boldsymbol{\mu}_p \cdot \hat{\mathbf{r}}) \hat{\mathbf{r}} - \boldsymbol{\mu}_p] = \mu_0 \mathbf{H}_{\text{dip}}, \quad (5.10)$$

**Thermal field.** Following Eq. (4) of Zhu et al. [12], write the sampled thermal field as

$$\mathbf{B}_{\text{th}} = \mu_0 \mathbf{H}_{\text{th}} = \mathcal{N}_1(0, u_B) \hat{\mathbf{x}} + \mathcal{N}_2(0, u_B) \hat{\mathbf{y}} + \mathcal{N}_3(0, u_B) \hat{\mathbf{z}}, \quad (5.11)$$

$$u_B = \sqrt{\frac{2\alpha k_B T}{V_{\text{FL}} M_s |\gamma| (1 + \alpha^2) \Delta t}}. \quad (5.12)$$

Here the  $\mathcal{N}_i(0, u_B)$  are independent normal random variables with zero mean and standard deviation  $u_B$ ,  $V_{\text{FL}}$  is the free-layer volume, and  $\Delta t$  is the duration over which a sampled field is held constant. With  $|\gamma|$  in  $\text{rad s}^{-1} \text{T}^{-1}$  and  $M_s$  in  $\text{A m}^{-1}$ ,  $u_B$  is in tesla; the corresponding  $\mathbf{H}$ -field standard deviation is  $u_H = u_B / \mu_0$ .

The source uses independent samples over  $\Delta t = 5 \text{ ps}$  [12]. When adopting another integration interval or LLG convention, recompute the noise amplitude consistently rather than reusing that numerical value.

**Total field.** The total one-sublattice field, with the stochastic field added after taking the energy derivative, is [17]

$$\mathbf{H}_{\text{eff}} = \mathbf{H}_{\text{ext}} + \mathbf{H}_{\text{an}} + \mathbf{H}_{\text{demag}} + \mathbf{H}_{\text{dip}} + \mathbf{H}_{\text{th}}. \quad (5.13)$$

Spin-transfer and spin-orbit torques are nonconservative and are not part of the energy derivative in Eq. (5.4).

**Implementation mapping.** The current one-sublattice `ODE_LLG_integral` implementation stores flux-density quantities in tesla [17]. Its `field_eta.m` evaluates the following expressions, where its input named `Hk` is  $B_k$ :

$$\begin{aligned} \mathbf{B}_{\text{an}} &= B_k m_y \hat{\mathbf{y}} \quad (\text{IMA}), & \mathbf{B}_{\text{an}} &= B_k m_z \hat{\mathbf{z}} \quad (\text{PMA}), \\ \mathbf{B}_{\text{demag}} &= -\mu_0 M_s \mathbf{N} \mathbf{m}, & \mathbf{B}_{\text{dip}} &= \mu_0 \mathbf{K}_{12} M_s \mathbf{m}_{\text{PL}}, \end{aligned}$$

after converting the input  $M_s$  from  $\text{emu cm}^{-3}$  to  $\text{A m}^{-1}$ . It sums these with the applied and thermal fields.

## 5.2.2 Two-Sublattice Collinear Model

**Inter-sublattice exchange field.** Let  $\mathbf{m}_1$  and  $\mathbf{m}_2$  be separately normalized sublattice magnetizations with saturation magnetizations  $M_{s,1}$  and  $M_{s,2}$ . The exchange-energy density convention used by the two-sublattice ferrimagnetic macrospin model of Yuan et al. [13] is

$$w_{\text{ex},12} = A_{12} \mathbf{m}_1 \cdot \mathbf{m}_2, \quad A_{12} > 0, \quad (5.14)$$

Applying  $\mathbf{H}_{\text{ex},i} = -(\mu_0 M_{s,i})^{-1} \partial w_{\text{ex},12} / \partial \mathbf{m}_i$  yields, for  $j \neq i$ ,

$$\mathbf{H}_{\text{ex},i} = -\frac{A_{12}}{\mu_0 M_{s,i}} \mathbf{m}_j, \quad \mathbf{B}_{\text{ex},i} = -\frac{A_{12}}{M_{s,i}} \mathbf{m}_j. \quad (5.15)$$

Only the exchange contribution unique to the two-sublattice model is stated here; the shared external, anisotropy, demagnetization, dipolar, and thermal field formulas are defined above and are not repeated. This two-sublattice exchange extension is publication-grounded and is not implemented by the `ODE_LLG_integral` repository.

## 6 Magnetization Dynamics

**LLGS form with field-like torque.** This macrospin equation is adapted from Eqs. (1)–(4) of Kong et al., with the atom index removed [3]. It is written in the flux-density convention:  $\mathbf{B}_{\text{eff}}$  and the torque amplitudes are in tesla and are paired with  $\gamma$  in  $\text{rad s}^{-1} \text{T}^{-1}$ .

$$\begin{aligned} \frac{d\mathbf{m}}{dt} = & -\gamma \mathbf{m} \times \mathbf{B}_{\text{eff}} + \alpha \mathbf{m} \times \frac{d\mathbf{m}}{dt} - \gamma \tau_{\text{DLT}} \mathbf{m} \times (\mathbf{m} \times \boldsymbol{\sigma}) \\ & + \gamma \tau_{\text{FLT}} \mathbf{m} \times \boldsymbol{\sigma}. \end{aligned} \quad (6.1)$$

**Equivalent LL form with field-like torque.** The algebraically equivalent form follows the same source convention [3].

$$\begin{aligned} (1 + \alpha^2) \frac{d\mathbf{m}}{dt} = & -\gamma \mathbf{m} \times \mathbf{B}_{\text{eff}} - \alpha \gamma \mathbf{m} \times (\mathbf{m} \times \mathbf{B}_{\text{eff}}) \\ & + \alpha \gamma \tau_{\text{DLT}} \mathbf{m} \times \boldsymbol{\sigma} - \gamma \tau_{\text{DLT}} \mathbf{m} \times (\mathbf{m} \times \boldsymbol{\sigma}) \\ & + \alpha \gamma \tau_{\text{FLT}} \mathbf{m} \times (\mathbf{m} \times \boldsymbol{\sigma}) + \gamma \tau_{\text{FLT}} \mathbf{m} \times \boldsymbol{\sigma}. \end{aligned} \quad (6.2)$$

**Torque amplitudes.** For a ferromagnetic layer [4],

$$\tau_{\text{DLT}} = \frac{\hbar \theta_{\text{SH}} J_c}{2e t_{\text{FM}} M_s}, \quad \tau_{\text{FLT}} = \beta \tau_{\text{DLT}}.$$

**Torque-sign convention.** In the convention of Eq. (6.1), the sign product of the damping-like and field-like SOT is negative when  $\beta > 0$  [1, 3]. Specifically, the scalar prefactors of their respective vector structures are  $-\gamma \tau_{\text{DLT}}$  and  $+\gamma \tau_{\text{FLT}}$ , so their sign product is

$$(-\gamma \tau_{\text{DLT}})(+\gamma \tau_{\text{FLT}}) = -\gamma^2 \beta \tau_{\text{DLT}}^2 < 0 \quad (\beta > 0).$$

Thus, a positive  $\beta$  means that the two SOT terms have oppositely signed scalar prefactors in the LLGS equation, even though the signed amplitudes  $\tau_{\text{DLT}}$  and  $\tau_{\text{FLT}}$  themselves have the same sign.

## 7 Thermal Stability

### 7.1 Thermal Stability Factor

**Equation.** For a single energy barrier  $E_b$ , the thermal stability factor is [19]

$$\Delta = \frac{K_u V}{k_B T} \quad (7.1)$$

**Units.**  $E_b$  and  $k_B T$  must use the same energy unit;  $\Delta$  is dimensionless.

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- [19] EE237 Week 10 lecture slides, MRAM Cells, Reliability, and Design Tradeoffs, slide 28.
- [20] EE237 Week 11 lecture slides, Spin-Orbit Torques and Switching Symmetry, slides 4–5.